

Seunguk Kang

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 Website  GitHub  LinkedIn

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Bachelor of Science in Chemistry

Daejeon, Republic of Korea
2022 – Present

Korea Science Academy (KSA)

High School Diploma in Chemistry

Busan, Republic of Korea
2019 – 2021

SKILLS

Programming and ML: Python, PyTorch, NumPy, Pandas, RDKit, Git/GitHub, Linux/Bash, HPC/Slurm; graph neural networks, equivariant neural networks, diffusion- and flow-based generative models, molecular generative modeling, transformer-based models, scientific machine learning.

Computational chemistry: DFT calculations (ORCA, ADF), molecular dynamics simulations (Amber), QM/MM simulations, molecular structure analysis and visualization (Mercury, ChemDraw, Chemcraft, Materials Studio, CSD).

Experimental and analytical: Powder X-ray diffraction (PXRD), SEM, EDS, TGA, UV-Vis, IR spectroscopy, surface area and porosimetry analysis, glove box handling, ligand purification, TLC, glass blowing for vacuum lines, electrode fabrication using ITO film.

RESEARCH EXPERIENCE

Intelligent Chemistry Lab, KAIST

Dec. 2025 – Present

Undergraduate Researcher Advisor: Prof. Woo Youn Kim

- Developing a neural free energy estimator (NFEF) for estimating free energy differences of small-molecule ensembles using learned nonequilibrium generative paths.
- Building AD-Agent, a mechanism-aware multi-agent framework for drug repurposing in Alzheimer's disease.
- Studied induced-fit antibody–antigen interaction modeling with a Riemannian Diffusion Bridge-based E(3)-equivariant generative framework.

Baik Group, KAIST

Jan. 2024 – Dec. 2025

Undergraduate Researcher Advisor: Prof. Mu-Hyun Baik

- Studied the mechanism of enantioselective amino-tetralin synthesis catalyzed by an indenyl-Rh complex.
- Performed DFT benchmarking studies on substituent-dependent reactivity in relation to Hammett substituent constants.
- Investigated the mechanism of a solid-state stereoselective [2+2] photocycloaddition.

Soft Material Assembly Group, KAIST

Jun. 2023 – Dec. 2023

Undergraduate Researcher Advisor: Prof. Dong Ki Yoon

- Studied tunable electrical conductivity anisotropy in a MOF–polymer composite using Cu₃(HHTP)₂.
- Investigated the applications of MOF–polymer composites as selective gas sensors.

Choi Group, KSA

Apr. 2019 – Dec. 2021

Student Researcher Advisor: Prof. Eun Young Choi

- Performed gas and ion exchange experiments on zeolite single crystals under high-vacuum conditions.
- Developed protocols to successfully activate fragile porphyrin-based metal–organic frameworks to improve their gas adsorption capacity.
- Led Singapore International Collaborative Research Program work on Zn-, Co-, and Ag-tartrate metal–organic frameworks and their antibacterial properties.

PUBLICATIONS

Gross, P.; Choi, H.; Pullara, W. A.; Im, H.; Ung, K.; **Kang, S.**; Baik, M.-H.; Blakey, S. B. Enantioselective Olefin 1,2-Arylation Catalyzed by a Planar Chiral Indenyl-Rhodium(III) Complex. *ACS Catal.* **2026**, 16(4), 3453–3463. doi:10.1021/acscatal.5c07589

Hong, T.; Lee, C.; Bak, Y.; Park, G.; Lee, H.; **Kang, S.**; Bae, T.-H.; Yoon, D. K.; Park, J. G. On-Demand Tunable Electrical Conductance Anisotropy in a MOF-Polymer Composite. *Small* **2024**, 20(18), 2309469. doi:10.1002/sml.202309469

SELECTED PRESENTATIONS

Spring Conference of the Korean Chemical Society	2025
Poster presentation, Chiral Indenyl Rh-Catalyzed Enantioselective Olefin 1,2-Arylation	
Research Presentation for Chemistry Department Workshop	2024
Oral presentation, Chiral Indenyl Rh-Catalyzed Enantioselective Olefin 1,2-Arylation	
Future Chemists Research Conference of the Korean Chemical Society	2021
Poster presentation, Biocompatible MOFs Developed by Facile Ultra-Fast Room Temperature Synthesis	
Thailand International Science Fair	2021
Oral presentation, Improvement of Gas Adsorption Efficiency of Fragile Porphyrin-based MOFs	
Autumn Conference of the Korean Polymer Society	2020
Poster presentation, Improvement of Gas Adsorption Efficiency of Fragile Porphyrin-based MOFs	
Japan Super Science Fair	2020
Oral presentation, Optimizing Synthesis Conditions of PPF-4: Effect of Acid Addition on Crystal Growth	

HONORS AND AWARDS

Best Participant Award	2024
Global Entrepreneurship Summer School	
2nd Prize	2024
IR Pitching Contest, Global Entrepreneurship Summer School	
Talent Award of Korea	2021
National Prize, Ministry of Education, Republic of Korea. Awarded to 100 individuals nationwide who exemplify future leadership.	
Meritorious Service Award	2021
Graduation of Korea Science Academy	
Academic Excellence Award and Comprehensive Studies Award	2020, 2021
Chemistry Department, Korea Science Academy; 1st and 2nd semesters	
Gold Prize	2017
World Invention Creativity Olympic (WICO), International Competition	
The Minister's Award	2017
Korean Youth Invention Idea Contest, Ministry of Science and ICT, Republic of Korea	
Silver Prize	2015
Youth Science Debate Competition, National Competition	

EXTRA ACTIVITIES

Team Leader (CEO)	2024
Mock start-up, KAIST ENablers	
Band Leader	2022 – 2023
KAIST undergraduate band, CarpeDiem	
Research Lead	2021
Singapore International Cooperative Research Team	
Band Leader	2019 – 2021
Korea Science Academy band, Loony	